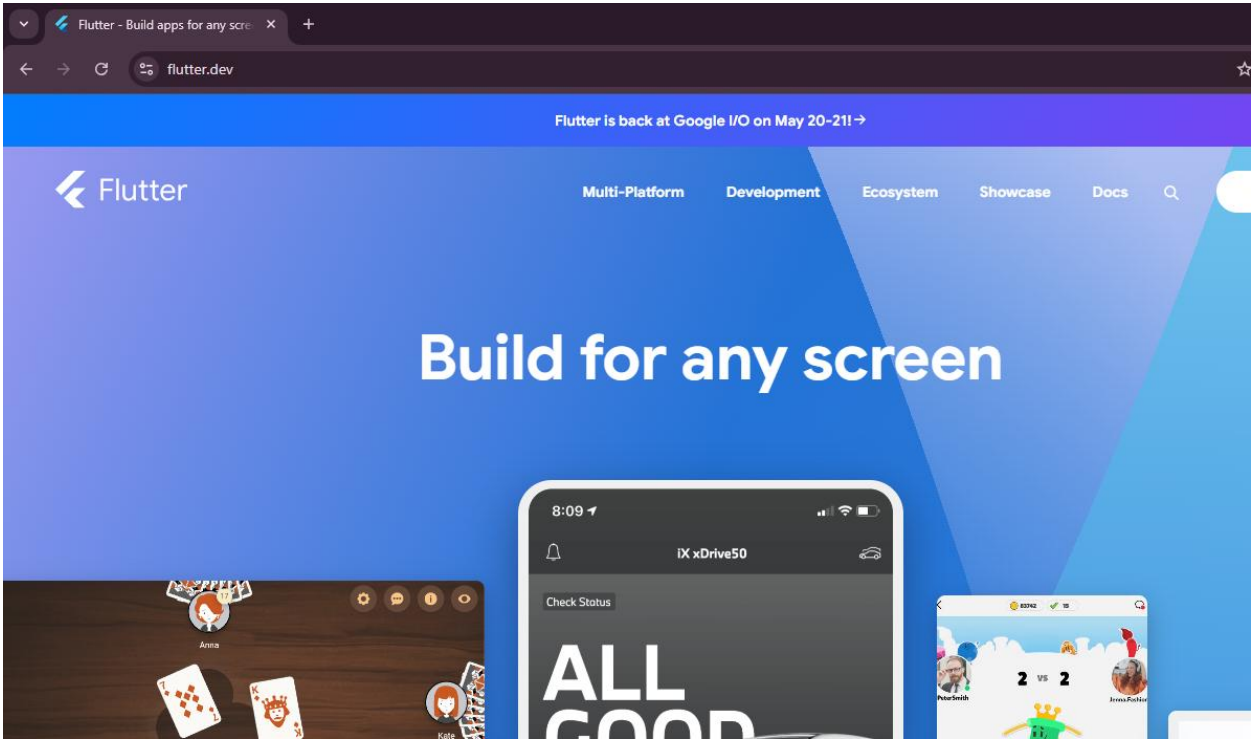
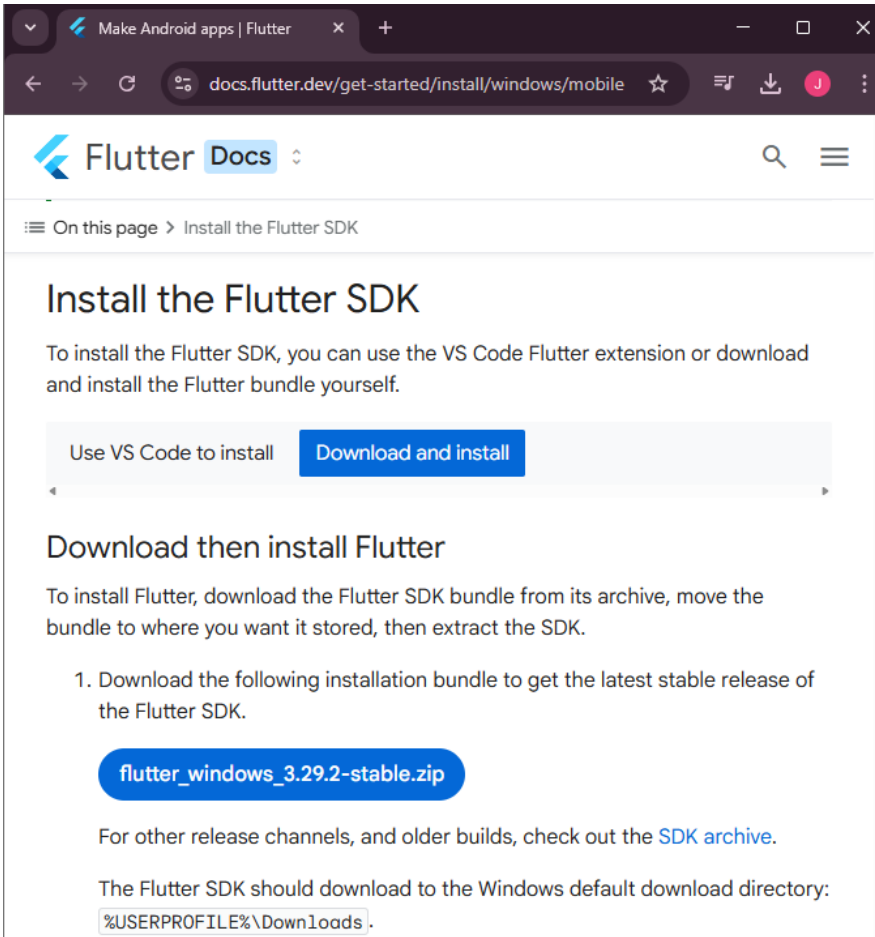


FLUTTER INSTALLATION GUIDE

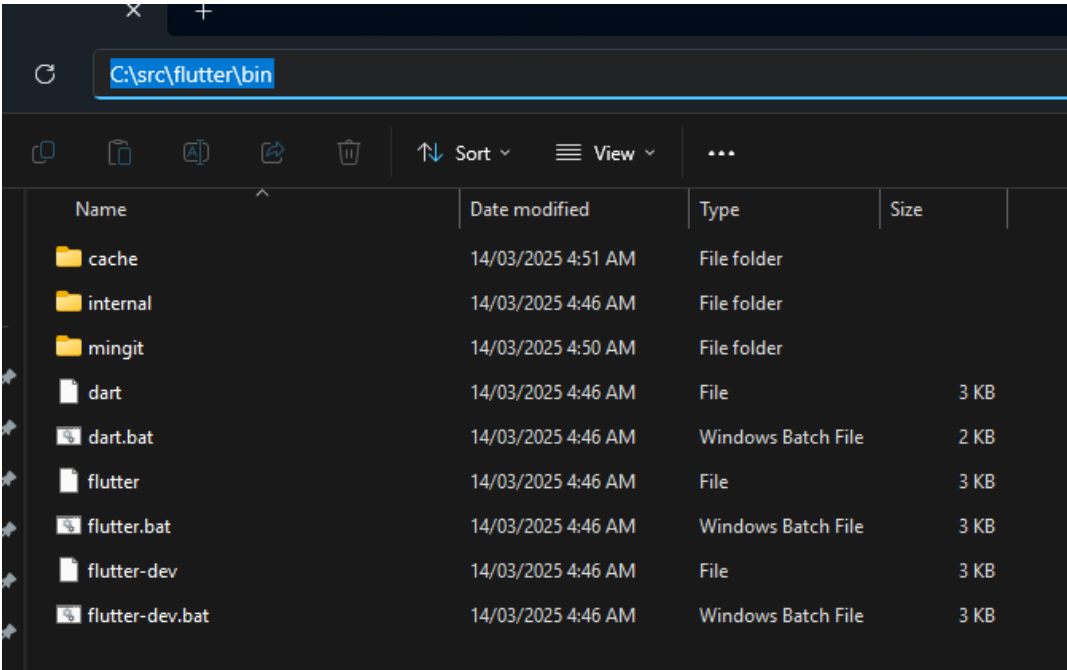
1. Go to flutter.dev and navigate to Docs.



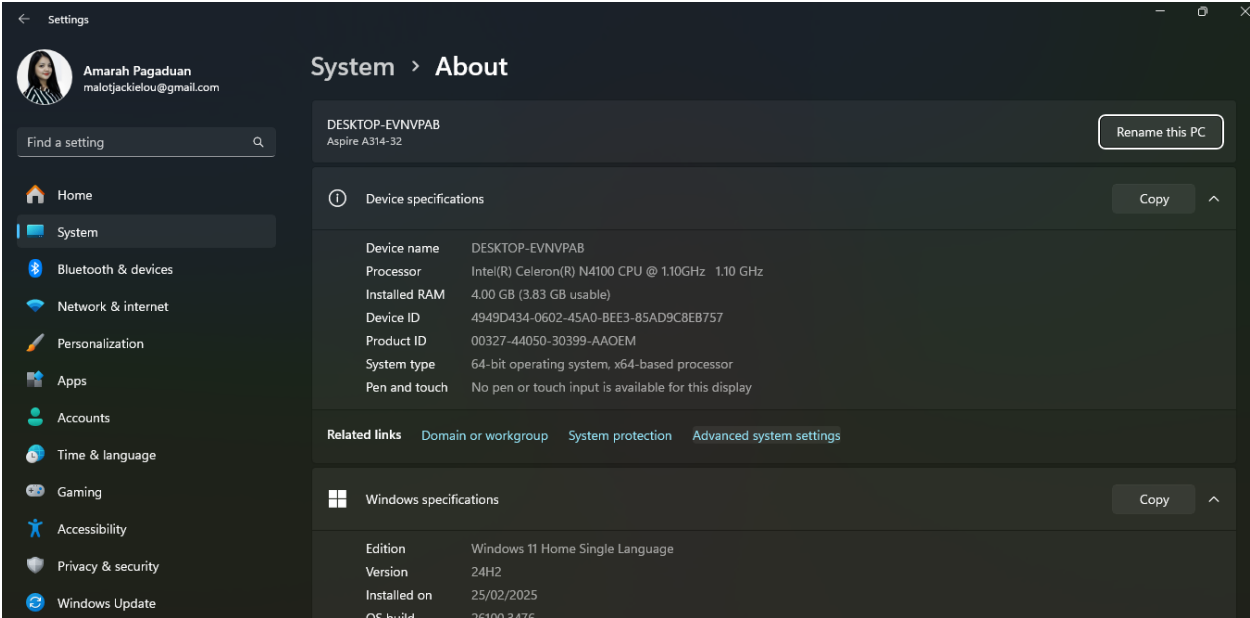
2. Click Get started, choose windows then select the platform we are to develop which is the Android. Scroll down and you can see the software and hardware minimum requirements.
3. Scroll further and you can see Install the Flutter SDK. Choose download and install then click on flutter_windows_3.29.2-stable.zip and wait for it to download.



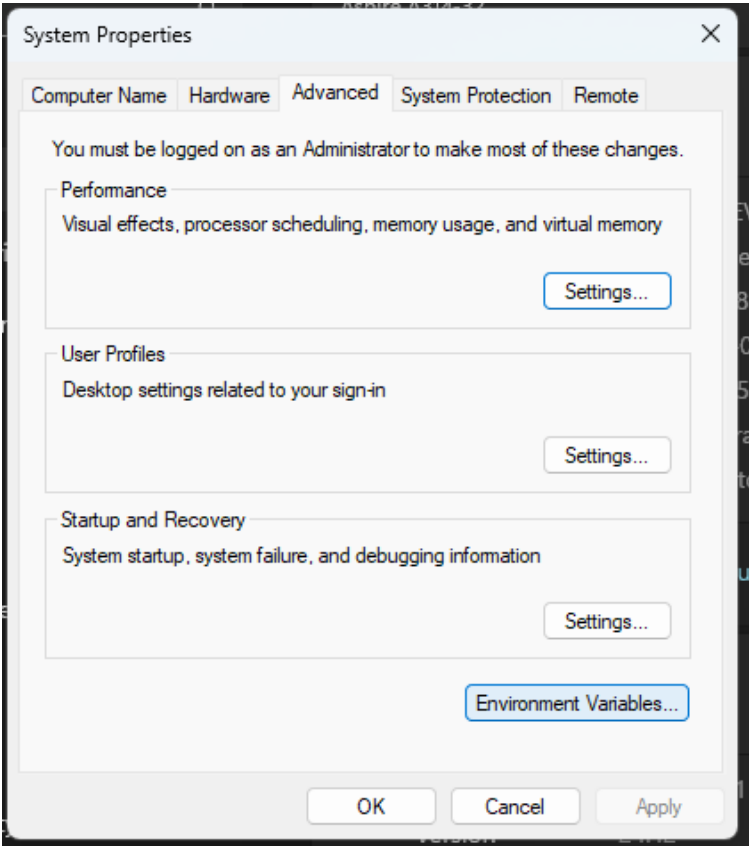
4. Now extract the downloaded file in C:\src\ then wait for it to extract.
5. After extracting, we need to update your windows path variable to run flutter commands in PowerShell. To do this, go to your flutter folder >bin then copy the file path.



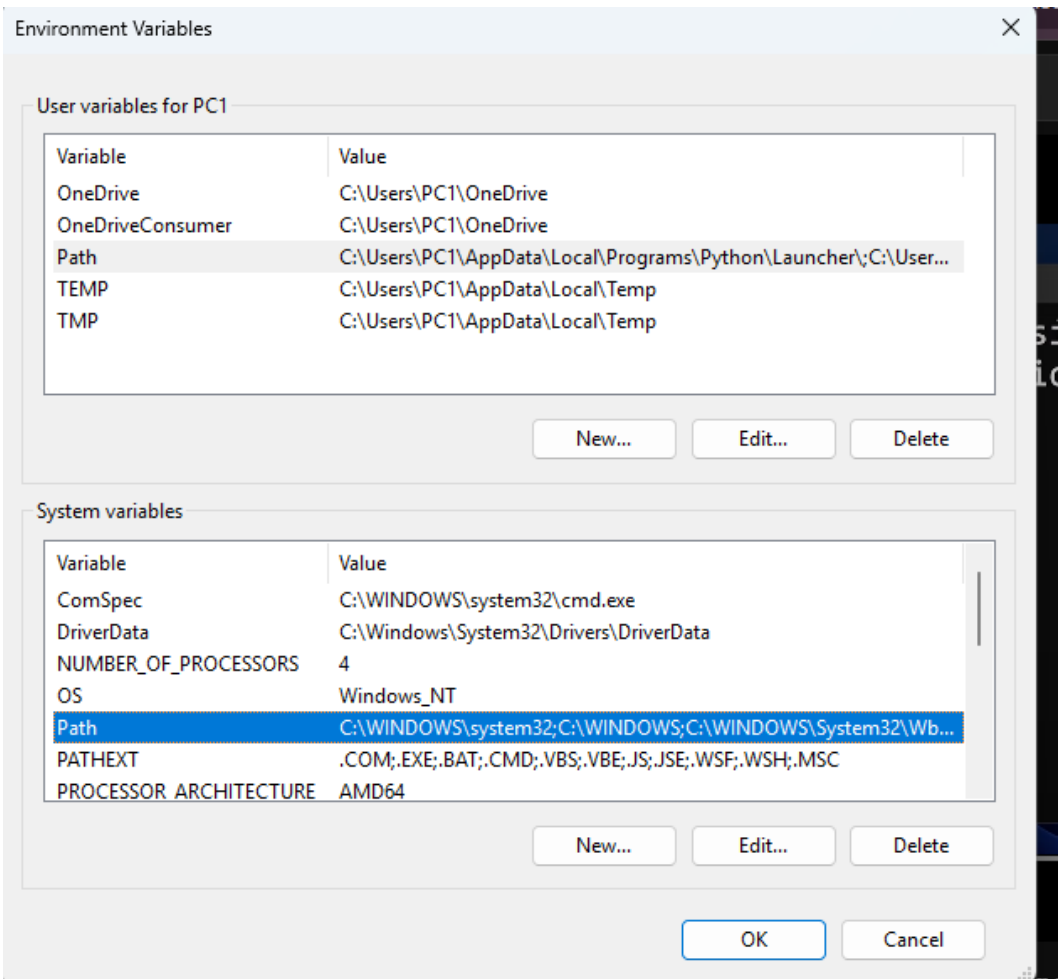
6. Now follow the steps provided in flutter.dev. so, press windows + pause in your keyboard and you will be directed to the system about as shown in the image below.



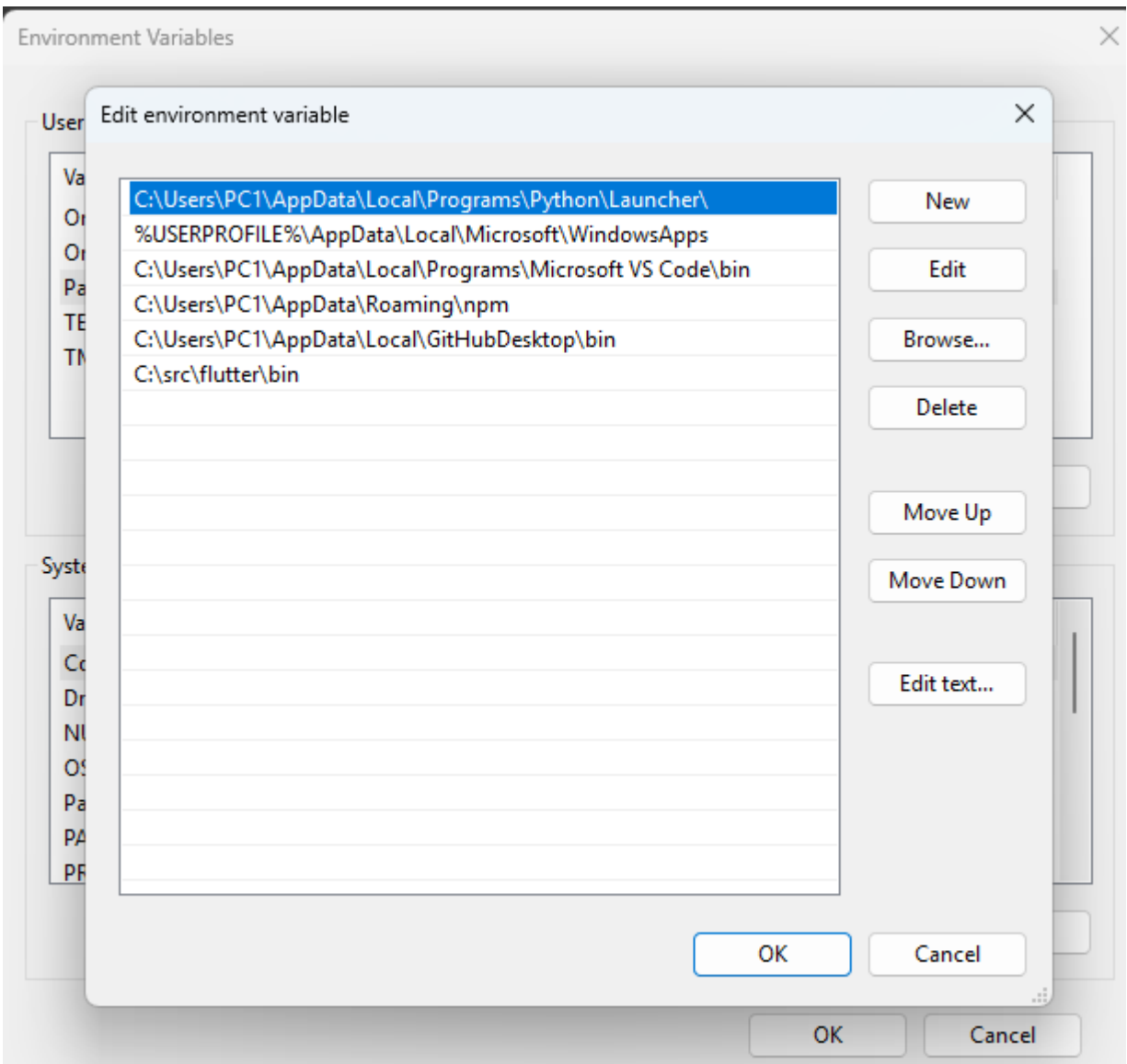
7. Click the advanced system settings > advanced>environment variables.



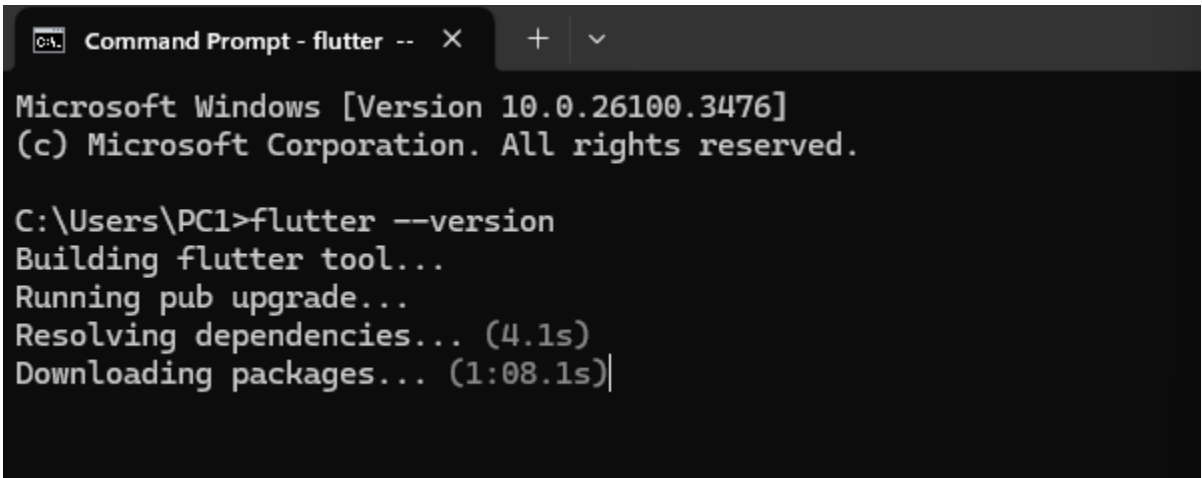
8. Locate Path in system variable and click edit.



9. Double clicks in an empty row and paste the flutter bin location. Click ok to exit.



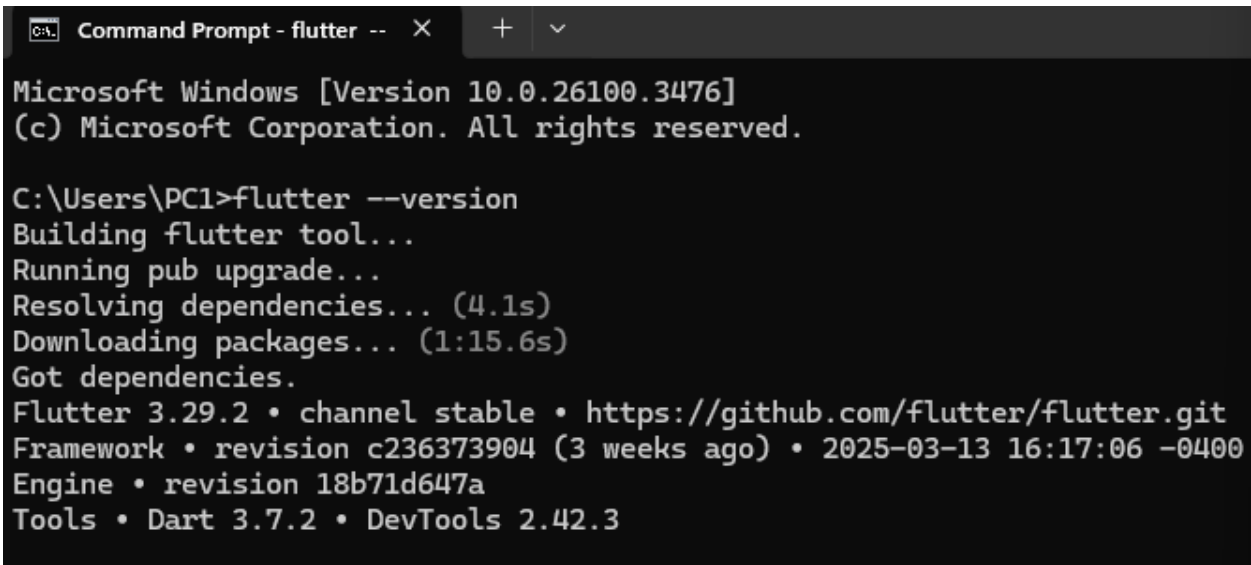
10. Now to enable these changes, close and open command prompts and PowerShell instances and type flutter --version. Wait for a few seconds to for it to download packages and resolve dependencies for smooth flow of mobile app development.



```
Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. All rights reserved.

C:\Users\PC1>flutter --version
Building flutter tool...
Running pub upgrade...
Resolving dependencies... (4.1s)
Downloading packages... (1:08.1s)
```

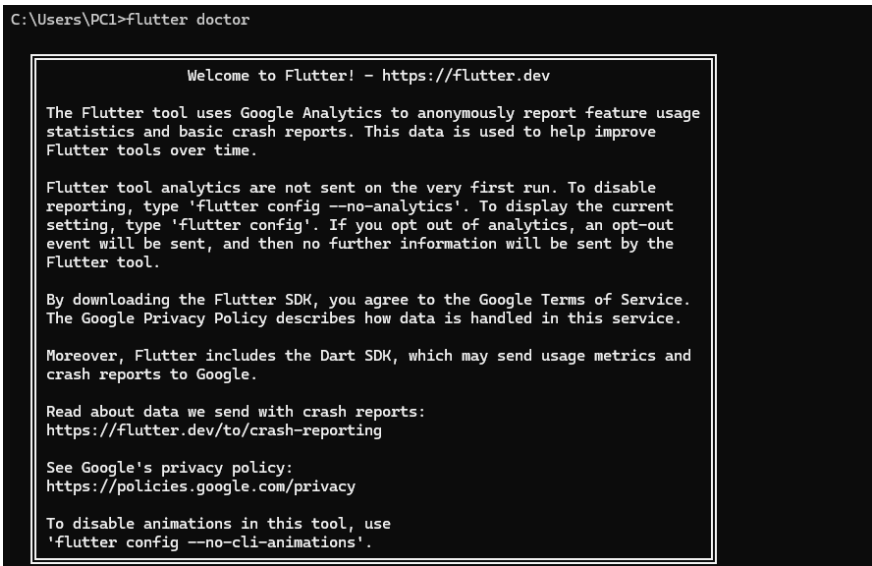
11. Wait for a moment until it displays the flutter version like the image below. In our case, the Flutter version is 3.29.2.



```
Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. All rights reserved.

C:\Users\PC1>flutter --version
Building flutter tool...
Running pub upgrade...
Resolving dependencies... (4.1s)
Downloading packages... (1:15.6s)
Got dependencies.
Flutter 3.29.2 • channel stable • https://github.com/flutter/flutter.git
Framework • revision c236373904 (3 weeks ago) • 2025-03-13 16:17:06 -0400
Engine • revision 18b71d647a
Tools • Dart 3.7.2 • DevTools 2.42.3
```

12. After knowing the flutter version, we can now install the IDE we are to use or configure our Android development by installing necessary SDK platform. Actually, there are 2 IDEs but preference aside, we will make use of Android Studio. Go ahead and install the Android Studio IDE. Follow the onscreen instruction. Wait for at least 2 minutes to install this IDE.
13. After installing, open cmd prompt again and type flutter doctor to see if there are any problems with the installation and to agree to the licenses of the SDK platform. In our case after typing flutter doctor, it displays the image below which is the agreement.



```
C:\Users\PC1>flutter doctor

Welcome to Flutter! - https://flutter.dev

The Flutter tool uses Google Analytics to anonymously report feature usage
statistics and basic crash reports. This data is used to help improve
Flutter tools over time.

Flutter tool analytics are not sent on the very first run. To disable
reporting, type 'flutter config --no-analytics'. To display the current
setting, type 'flutter config'. If you opt out of analytics, an opt-out
event will be sent, and then no further information will be sent by the
Flutter tool.

By downloading the Flutter SDK, you agree to the Google Terms of Service.
The Google Privacy Policy describes how data is handled in this service.

Moreover, Flutter includes the Dart SDK, which may send usage metrics and
crash reports to Google.

Read about data we send with crash reports:
https://flutter.dev/to/crash-reporting

See Google's privacy policy:
https://policies.google.com/privacy

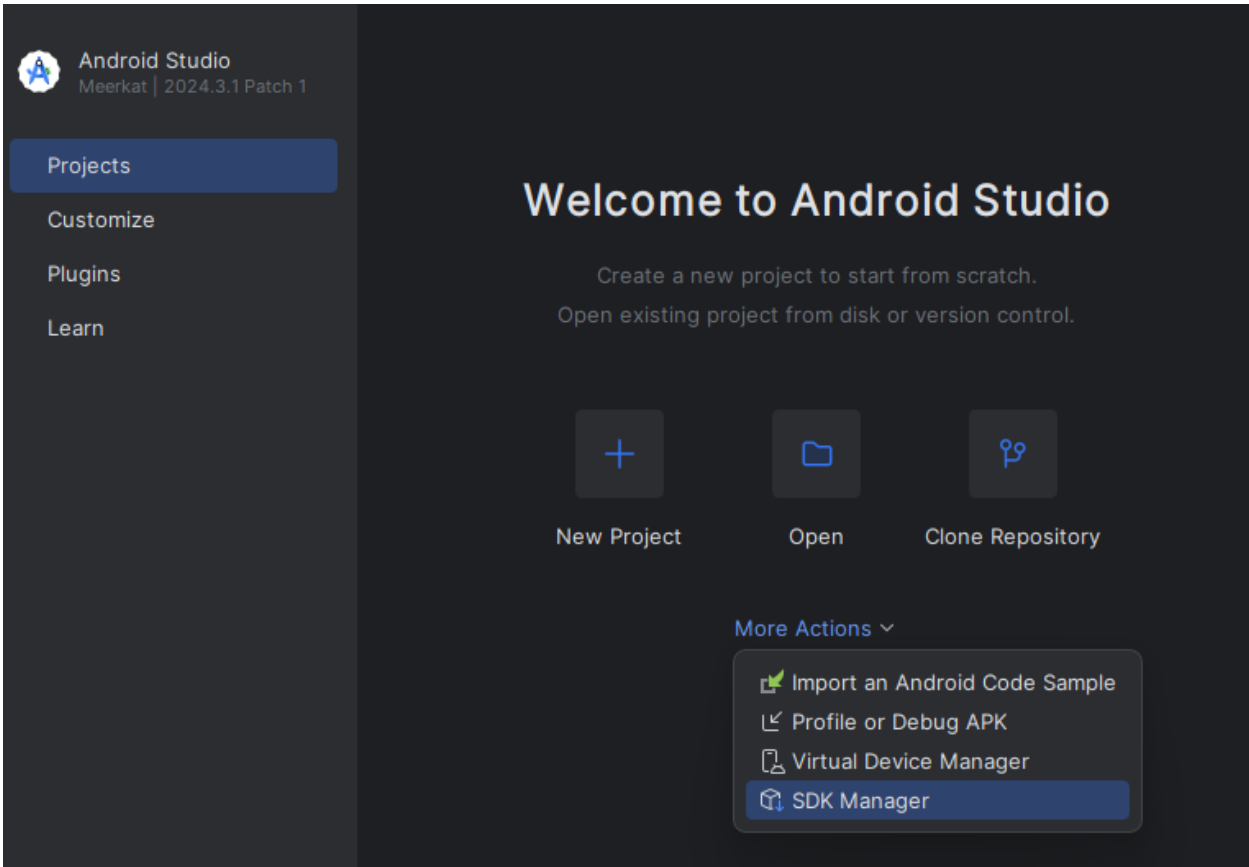
To disable animations in this tool, use
'flutter config --no-cli-animations'.
```

14. And following the doctor summary. As noticed, Doctor found issues in 2 categories which is marked as x.

```
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.29.2, on Microsoft Windows [Version 10.0.26100.3476])
[✓] Windows Version (11 Home Single Language 64-bit, 24H2, 2009)
[!] Android toolchain - develop for Android devices (Android SDK version 34.0.0)
    X cmdline-tools component is missing
      Run 'path/to/sdkmanager --install "cmdline-tools;latest"'
      See https://developer.android.com/studio/command-line for more details.
    X Android license status unknown.
      Run 'flutter doctor --android-licenses' to accept the SDK licenses.
      See https://flutter.dev/to/windows-android-setup for more details.
[✓] Chrome - develop for the web
[X] Visual Studio - develop Windows apps
    X Visual Studio not installed; this is necessary to develop Windows apps.
      Download at https://visualstudio.microsoft.com/downloads/.
      Please install the "Desktop development with C++" workload, including all of
[✓] Android Studio (version 2022.3)
[✓] VS Code (version 1.98.2)
[✓] Connected device (3 available)
[✓] Network resources

! Doctor found issues in 2 categories.
```

15. Now to solve the cmdline-tools component is missing, run android sdk and click more actions then select SDK manager.



16. Under SDK Tools, click Android SDK Command-line tools (latest), click Apply > OK and wait to complete the requested actions. This will take 3 minutes to complete. Just click finish and OK. Now let's try to run flutter doctor again in cmd.

17. Now let's try to solve the android licenses not accepted by simply copy and pasting flutter doctor --android-licenses. Wait for a few seconds and just type y until complete. Now after ALL SDK licenses accepted try to run again flutter doctor to check if there are missing dependencies so we can solve it to have a smooth flow of development.

18. Based from the summary, it shows that visual studio not installed. Download and install it. Follow the onscreen instruction. Take note of this: Please install the "Desktop development with C++" workload, including all of its default components. This installation will take a while so you can get your coffee first.

19. After installing, run again flutter doctor to check if we successfully install it all. Your doctor summary should indicate no issues found as illustrated in the image below.

```
Command Prompt - flutter d
Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. All rights reserved.

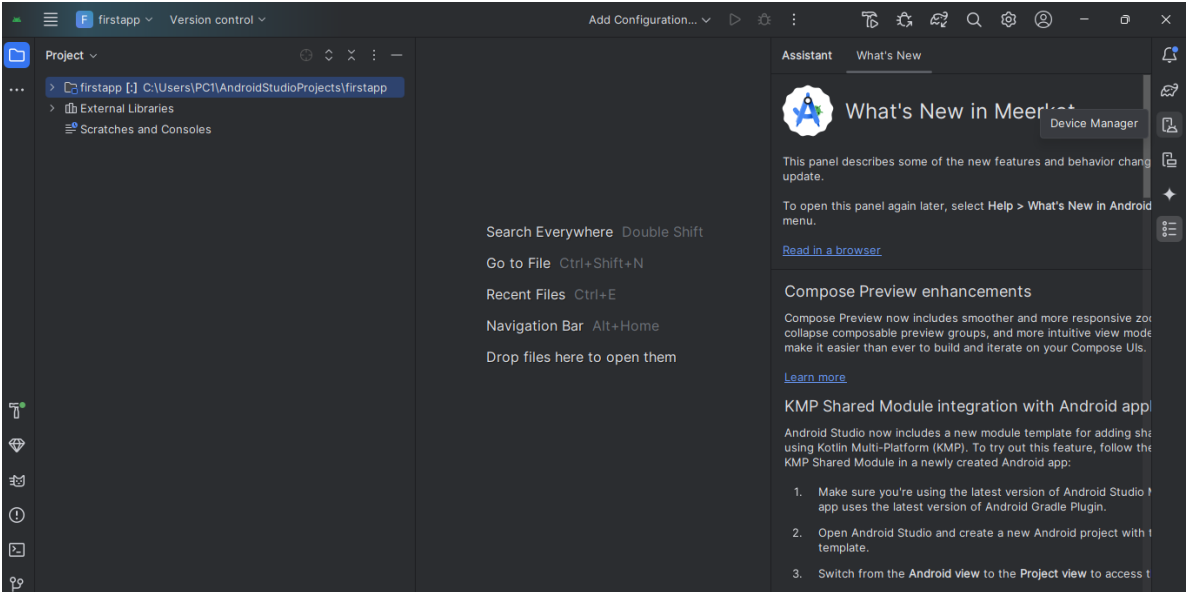
C:\Users\PC1>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.29.2, on Microsoft Windows [Version 10.0.26100.3476], locale en-PH)
[✓] Windows Version (11 Home Single Language 64-bit, 24H2, 2009)
[✓] Android toolchain - develop for Android devices (Android SDK version 34.0.0)
[✓] Chrome - develop for the web
[✓] Visual Studio - develop Windows apps (Visual Studio Community 2022 17.14.0 Preview 2.0)
[✓] Android Studio (version 2024.3)
[✓] VS Code (version 1.99.0)
[✓] Connected device (3 available)
[✓] Network resources

• No issues found!
```

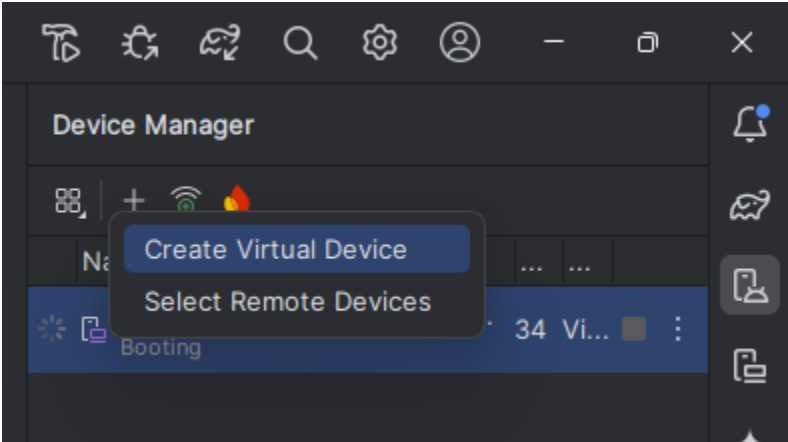
CONFIGURE YOUR TARGET ANDROID DEVICE

After successful installation, we need to set up our android emulator. Now how to do that?

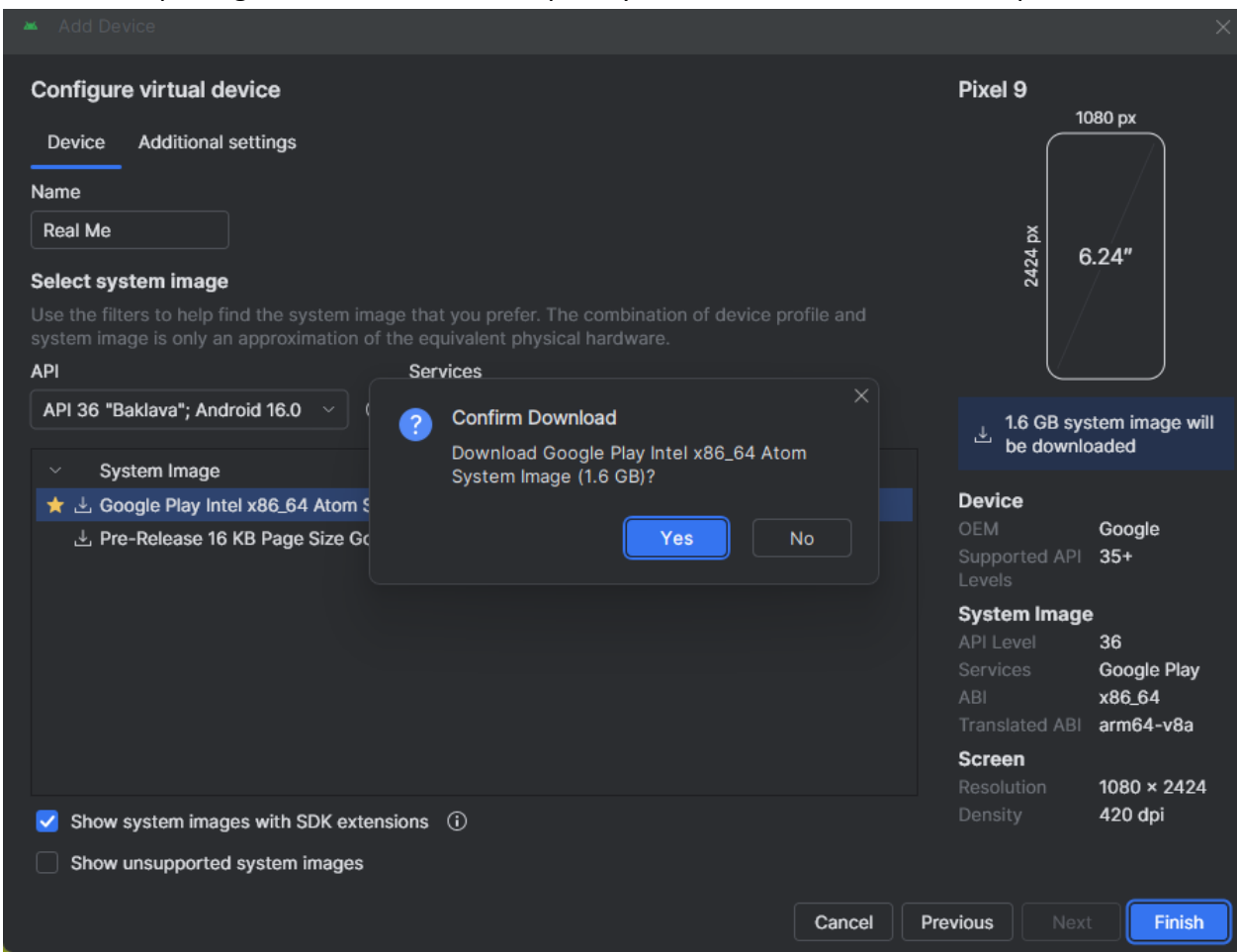
- 1. Enable the VM acceleration on your development computer by starting your android studio.
- 2. Run your android studio then create new project. Name it firstapp then hit finish. Wait until requested actions are complete. This will take a few seconds or depends on your internet connection. Once done, click finish. Wait for it to generate sources.
- 3. After generating sources, you will be directed to the page below. At the right side if you will hover your mouse, you will find the Device manager as shown in the image below.



- 4. Click the device manager and click Virtual. Wait for it to load the devices.



- 5. Phone category is selected, so just select the name of your device for example let's say our device is Pixel 9. Select it and click Next. You can rename the device. Check the show system images with SDK extensions. Then just click finish.
- 6. When you hit finish, it will provide a prompt asking to confirm download. Hit yes and wait for the packages to be download completely. This will take a minute so be patient.



- 7. Once done, click finish. Wait for it to create ADV. Wait for it to import necessary libraries, scan files, update indexes and repository data. Wait for it to launch the emulator once done.